

1) Consider the Campbell and Shiller (1988) log-linear (approximate) present value identity for equities
(h is in years):

$$d_{P_t} = \frac{\kappa_0 \cdot (e^{\kappa} - 1)}{1 - \kappa} + \sum_{h=1}^H e^{h-1} \cdot r_{t+h} - \sum_{h=1}^H e^{h-1} \cdot \Delta d_{t+h} + e^H \cdot d_{P_{t+H}} \quad (1.1)$$

$$= \frac{\kappa_0 \cdot (e^{\kappa} - 1)}{1 - \kappa} + \sum_{h=1}^H e^{h-1} \cdot E_t[r_{t+h}|A] - \sum_{h=1}^H e^{h-1} \cdot E_t[\Delta d_{t+h}|A] + e^H \cdot E_t[d_{P_{t+H}}|A] \quad (1.2)$$

$$= \frac{\kappa_0}{1 - \kappa} + \sum_{h=1}^H e^{h-1} \cdot E_t[r_{t+h}|A] - \sum_{h=1}^H e^{h-1} \cdot E_t[\Delta d_{t+h}|A] \quad (1.3)$$

where $\kappa = 1/(1 + e^{\kappa})$ and $\kappa_0 = -\log(\kappa) = (1 - \kappa) \cdot \log(1/\kappa - 1)$.

Starting from $R_{t,t+1} = (P_{t+1} + D_{t+1})/P_t$, derive the (approximate) Equations 1.1, 1.2, and 1.3. Explain and justify any assumptions you make.

First, assume $D > 0$

$$R_{t,t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}$$

$$\log(R_{t,t+1}) = \log\left(\frac{P_{t+1} + D_{t+1}}{P_t}\right)$$

$$\log(R_{t,t+1}) = \log\left(\frac{P_{t+1} + D_{t+1}}{P_{t+1}} \cdot P_{t+1}\right) = \log(P_{t+1}) + \log\left(1 + \frac{D_{t+1}}{P_{t+1}}\right) - \log(P_t)$$

$$= \log(P_{t+1}) + \log\left(1 + \frac{D_{t+1}}{P_{t+1}}\right) - \log(P_t)$$

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$$\text{def } Y_{t,t+1} = \log(R_{t,t+1}) \quad P_t = \log(P_t) \quad d_t = \log(d_t)$$

$$\text{given that } \frac{D_{t+1}}{P_{t+1}} = e^{\log\left(\frac{D_{t+1}}{P_{t+1}}\right)} = e^{d_{t+1} - P_{t+1}}$$

$$\log(R_{t,t+1}) = \log(P_{t+1}) + \log\left(1 + \frac{D_{t+1}}{P_{t+1}}\right) - \log(P_t)$$

$$Y_{t,t+1} = P_{t+1} + \log(1 + e^{d_{t+1} - P_{t+1}}) - P_t$$

$$\text{def } f(x) \equiv \log(1 + e^x), \text{ where } x = d_{t+1} - P_{t+1}$$

approximates this function around the average log dividend price ratio $\bar{x} = \bar{d} - \bar{p}$

From first order Taylor approximation $f(x) \approx f(\bar{x}) + f'(\bar{x}) \cdot (x - \bar{x})$

$$f(d_{t+1} - P_{t+1}) \approx f(\bar{d} - \bar{p}) + f'(\bar{d} - \bar{p}) \cdot ((d_{t+1} - P_{t+1}) - (\bar{d} - \bar{p}))$$

$$\text{We have } f(x) = \log(1 + e^x)$$

$$\text{thus } f'(x) = \frac{e^x}{1 + e^x}$$

$$f'(\bar{d} - \bar{p}) = \frac{e^{\bar{d} - \bar{p}}}{1 + e^{\bar{d} - \bar{p}}}$$

$$\text{def } k \equiv \frac{1}{1 + e^{\bar{d} - \bar{p}}}$$

$$\text{then } 1 - k = \frac{e^{\bar{d} - \bar{p}}}{1 + e^{\bar{d} - \bar{p}}}$$

$$\text{so } f'(\bar{d} - \bar{p}) = 1 - k$$

$$\text{Hence, } f(d_{t+1} - P_{t+1}) \approx f(\bar{d} - \bar{p}) + f'(\bar{d} - \bar{p}) \cdot ((d_{t+1} - P_{t+1}) - (\bar{d} - \bar{p}))$$

can be written as

$$f(d_{t+1} - P_{t+1}) \approx f(\bar{d} - \bar{p}) + (1 - k) \cdot ((d_{t+1} - P_{t+1}) - (\bar{d} - \bar{p}))$$

$$= f(\bar{d} - \bar{p}) - (1 - k)(\bar{d} - \bar{p}) + (1 - k)(d_{t+1} - P_{t+1})$$

$$\text{def } k_0 \equiv f(\bar{d} - \bar{p}) - (1 - k)(\bar{d} - \bar{p}), k_0 \text{ is a constant}$$

$$\text{since } f(\bar{d} - \bar{p}) = \log(1 + e^{\bar{d} - \bar{p}}), \text{ and } k = \frac{1}{1 + e^{\bar{d} - \bar{p}}}$$

$$\text{then } 1 + e^{\bar{d} - \bar{p}} = \frac{1}{k}$$

$$\text{Hence, } f(\bar{d} - \bar{p}) = \log(1 + e^{\bar{d} - \bar{p}}) = \log\left(\frac{1}{k}\right) = -\log k$$

$$\text{and } e^{\bar{d} - \bar{p}} = \frac{1}{k} - 1$$

$$\bar{d} - \bar{p} = \log\left(\frac{1}{k} - 1\right)$$

$$k_0 \equiv f(\bar{d} - \bar{p}) - (1 - k)(\bar{d} - \bar{p})$$

$$= -\log k - (1 - k) \log\left(\frac{1}{k} - 1\right)$$

$$\text{We have } \log(1 + e^{d_{t+1} - P_{t+1}}) = f(d_{t+1} - P_{t+1}) \approx k_0 + (1 - k)(d_{t+1} - P_{t+1})$$

$$\log(1 + e^{d_{t+1} - P_{t+1}}) \approx k_0 + (1 - k)(d_{t+1} - P_{t+1})$$

$$\text{given that } Y_{t,t+1} = P_{t+1} + \log(1 + e^{d_{t+1} - P_{t+1}}) - P_t$$

$$\approx P_{t+1} + k_0 + (1 - k)(d_{t+1} - P_{t+1}) - P_t$$

$$= k_0 + (1 - k)d_{t+1} + kP_{t+1} - P_t$$

$$= k_0 + (1 - k)d_{t+1} - d_t + kP_{t+1} - P_t + d_t$$

$$= k_0 + (d_{t+1} - d_t) + k(P_{t+1} - d_{t+1}) - (P_t - d_t)$$

$$= k_0 + \Delta d_{t+1} + k(P_{t+1} - d_{t+1}) - (P_t - d_t)$$

Therefore, we have

$$Y_{e,t+1} \approx K_0 + \Delta d_{t+1} + K(P_{t+1} - d_{t+1}) - (P_t - d_t) \quad \text{w/} \quad k = \frac{1}{1+e^{\frac{1}{1-k}}} \quad \& \quad K_0 = -\log(k) - (1-k) \cdot \log\left(\frac{1}{k} - 1\right)$$

Isolating $P_t - d_t$ and recursively substituting $P_{t+h} - d_{t+h}$, we have

$$\begin{aligned} (P_t - d_t) &\approx K_0 - Y_{e,t+1} + \Delta d_{t+1} + K(P_{t+1} - d_{t+1}) \\ &= K_0 - Y_{e,t+1} + \Delta d_{t+1} + K[K_0 - Y_{e,t+2} + \Delta d_{t+2} + K(P_{t+2} - d_{t+2})] \\ &= (1+K)K_0 + (\Delta d_{t+1} + K\Delta d_{t+2}) - (Y_{e,t+1} + KY_{e,t+2}) + K^2(P_{t+2} - d_{t+2}) \\ &\vdots \quad \text{keep substituting } P_{t+h} - d_{t+h} \\ &= \frac{K_0(1-K^N)}{1-K} + \sum_{h=1}^N K^{h-1} \Delta d_{t+h} - \sum_{h=1}^N K^{h-1} Y_{e,t+h} + K^N(P_{t+N} - d_{t+N}) \end{aligned}$$

Thus, we have

$$d_t - P_t \approx \frac{K_0(K^N - 1)}{1-K} + \sum_{h=1}^N K^{h-1} Y_{e,t+h} - \sum_{h=1}^N K^{h-1} \Delta d_{t+h} + K^N(d_{t+N} - P_{t+N})$$

$$dP_t \approx \frac{K_0(K^N - 1)}{1-K} + \sum_{h=1}^N K^{h-1} Y_{e,t+h} - \sum_{h=1}^N K^{h-1} \Delta d_{t+h} + K^N dP_{t+N}$$

Take expectation at t , we have

$$dP_t = \frac{K_0(K^N - 1)}{1-K} + \sum_{h=1}^N K^{h-1} E_t[Y_{e,t+h}] - \sum_{h=1}^N K^{h-1} E_t[\Delta d_{t+h}] + K^N E_t[dP_{t+N}]$$

Suppose $N \rightarrow \infty$, given that $k = \frac{1}{1+e^{\frac{1}{1-k}}} \in (0, 1)$, $\lim_{N \rightarrow \infty} K^N = 0$

And because of the no-bubble condition, $\lim_{N \rightarrow \infty} K^N E_t[dP_{t+N}] = 0$

we have

$$dP_t = \frac{-K_0}{1-K} + \sum_{h=1}^{\infty} K^{h-1} E_t[Y_{e,t+h}] - \sum_{h=1}^{\infty} K^{h-1} E_t[\Delta d_{t+h}]$$